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1 Problem

2 Data

3 Solution

- Solving Conics using Matrices
- Plots

4 C Code

5 Python Code

Problem Statement

Question : Consider the parabola $y^2 = 4x$. Let **S** be the focus of the parabola. A pair of tangents drawn to the parabola from the point **R** = $(-2,1)$ meet the parabola at **R₁** and **R₂** . Let **Q₁** and **Q₂** be points on the lines SR_1 and SR_2 respectively such that RQ_1 is perpendicular to SR_1 and RQ_2 is perpendicular to SR_2 . Then, which of the following is/are TRUE ?

① $SQ_1 = 2$

② $Q_1Q_2 = \frac{3\sqrt{10}}{5}$

③ $RQ_1 = 3$

④ $SQ_2 = 1$

Data

Name	Value
Parabola	$\mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - \begin{pmatrix} 4 & 0 \end{pmatrix} \mathbf{x} = 0$

Table: Conic representation

Solving Conics using Matrices

On comparing Parabola with

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (4.1)$$

The parameters of the Parabola are

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}, f = 0 \quad (4.2)$$

As $\mathbf{V}$ is an upper triangular matrix, we get the eigen values as the diagonal entries

$$\lambda_1 = 0 \qquad \lambda_2 = 1 \quad (4.3)$$

For a Symmetric Matrix, we have the Eigen decomposition

$$\mathbf{V} = \mathbf{P} \mathbf{D} \mathbf{P}^T \quad (4.4)$$

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{V} \quad (4.5)$$

$$\mathbf{P} = \begin{pmatrix} \mathbf{p}_1 & \mathbf{p}_2 \end{pmatrix} \text{ and } \mathbf{P}^\top \mathbf{P} = \mathbf{I} \quad (4.6)$$

Solving both

$$\mathbf{V} = \mathbf{PVP}^\top \text{ and } \mathbf{P}^\top \mathbf{P} = \mathbf{I} \quad (4.7)$$

We get

$$\mathbf{P} = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix} \quad (4.8)$$

$$\mathbf{p}_1 = \begin{pmatrix} \pm 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{p}_2 = \begin{pmatrix} 0 \\ \pm 1 \end{pmatrix} \quad (4.9)$$

For a parabola, Eccentricity = $e = 1$

$$\mathbf{n} = \sqrt{\lambda_2} \mathbf{p}_1 = \mathbf{p}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.10)$$

$$c = \frac{\|\mathbf{u}\|^2 - \lambda_2 f}{2\mathbf{u}^\top \mathbf{n}} = \frac{4}{-4} = -1 \quad (4.11)$$

$$\text{Focus} = \mathbf{F} = \mathbf{S} = \frac{ce^2 \mathbf{n} - \mathbf{u}}{\lambda_2} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.12)$$

$$\mathbf{R} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad (4.13)$$

Given, The point of contact $\mathbf{q}$ the equation of tangent to (4.1) is

$$(\mathbf{V}\mathbf{q} + \mathbf{u})^\top \mathbf{x} + \mathbf{u}^\top \mathbf{q} + f = 0 \quad (4.14)$$

As the tangent passes through $\mathbf{R}$

$$(\mathbf{V}\mathbf{q} + \mathbf{u})^\top \mathbf{R} + \mathbf{u}^\top \mathbf{q} + f = 0 \quad (4.15)$$

Also $\mathbf{q}$ lies on the conic

$$\mathbf{q}^\top \mathbf{V}\mathbf{q} + 2\mathbf{u}^\top \mathbf{q} = 0 \quad (4.16)$$

$$\mathbf{V}^\top = \mathbf{V}, (\mathbf{V}\mathbf{q} + \mathbf{u})^\top \mathbf{R} + \mathbf{u}^\top \mathbf{q} = \mathbf{q}^\top \mathbf{V}\mathbf{R} + \mathbf{u}^\top \mathbf{R} + \mathbf{u}^\top \mathbf{q} = 0 \quad (4.17)$$

$$\mathbf{V}\mathbf{R} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } \mathbf{V}\mathbf{R} + \mathbf{u} = \mathbf{R} \text{ and } \mathbf{u}^\top \mathbf{R} = 4 \quad (4.18)$$

Also $\mathbf{q}^\top \mathbf{VR}$, $\mathbf{R}^\top \mathbf{Vq}$ are scalars. For a scalar the transpose of a scalar is the same scalar. So

$$\mathbf{q}^\top \mathbf{VR} = (\mathbf{q}^\top \mathbf{VR})^\top = \mathbf{R}^\top \mathbf{Vq} \quad (4.19)$$

From (4.17)

$$\mathbf{q}^\top \mathbf{VR} + 4 + (\mathbf{R} - \mathbf{VR})^\top \mathbf{q} = 0 \quad (4.20)$$

$$\mathbf{q}^\top \mathbf{VR} + 4 + \mathbf{R}^\top \mathbf{q} - \mathbf{R}^\top \mathbf{Vq} = 0 \quad (4.21)$$

$$\mathbf{R}^\top \mathbf{q} = -4 \quad (4.22)$$

$\mathbf{q}$ can be expressed as combination of $\mathbf{R}$ and its orthogonal component

$$\mathbf{q} = \lambda \mathbf{R} + \mathbf{z}, \text{ where } \mathbf{R}^\top \mathbf{z} = 0 \quad (4.23)$$

$$\mathbf{R}^\top \mathbf{q} = \lambda \mathbf{R}^\top \mathbf{R} = -4 \quad (4.24)$$

$$\lambda = \frac{-4}{5} \quad (4.25)$$

Substituting (4.23) in (4.16)

$$\lambda^2 \mathbf{R}^\top \mathbf{VR} + 2\lambda \mathbf{R}^\top \mathbf{Vz} + \mathbf{z}^\top \mathbf{Vz} + 8\lambda + 2\mathbf{u}^\top \mathbf{z} = 0 \quad (4.26)$$

$$\mathbf{R}^\top \mathbf{V} \mathbf{R} = 1 \text{ and } \lambda = \frac{-4}{5} \quad (4.27)$$

Substituting these and simplifying the equation

$$25\mathbf{z}^\top \mathbf{V} \mathbf{z} - 40\mathbf{R}^\top \mathbf{V} \mathbf{z} + 50\mathbf{u}^\top \mathbf{z} - 144 = 0 \quad (4.28)$$

$$\mathbf{z} = \begin{pmatrix} \frac{-\mathbf{u}^\top \mathbf{z}}{2} \\ \mathbf{R}^\top \mathbf{V} \mathbf{z} \end{pmatrix} \quad (4.29)$$

$$\mathbf{R}^\top \mathbf{z} = 0 \implies \mathbf{R}^\top \mathbf{V} \mathbf{z} = -\mathbf{u}^\top \mathbf{z} \quad (4.30)$$

Let

$$\mathbf{R}^\top \mathbf{V} \mathbf{z} = -\mathbf{u}^\top \mathbf{z} = k \quad (4.31)$$

Then

$$\mathbf{z}^\top \mathbf{V} \mathbf{z} = \left(\mathbf{R}^\top \mathbf{V} \mathbf{z} \right)^2 = k^2 \quad (4.32)$$

Then from (4.28)

$$25k^2 - 90k - 144 = 0 \quad (4.33)$$

$$\left(k - \frac{24}{5}\right) \left(k + \frac{6}{5}\right) = 0 \quad (4.34)$$

$$k = \frac{24}{5} \text{ or } k = \frac{-6}{5} \quad (4.35)$$

$$\mathbf{z} = \frac{1}{5} \begin{pmatrix} 12 \\ 24 \end{pmatrix} \text{ or } \mathbf{z} = \frac{1}{5} \begin{pmatrix} -3 \\ -6 \end{pmatrix} \quad (4.36)$$

$$\mathbf{q} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \text{ or } \mathbf{q} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (4.37)$$

As $\mathbf{R}_1, \mathbf{R}_2$ are the point of contact of Tangents $\{\mathbf{R}_1, \mathbf{R}_2\} \in \mathbf{q}$

$$\mathbf{R}_1 = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \text{ and } \mathbf{R}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (4.38)$$

If $\mathbf{A}, \mathbf{B}$ are on line $\mathbf{h}^\top \mathbf{x} = d$ then

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}^\top \mathbf{h} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.39)$$

For the line passing through $\mathbf{R}_1$ and $\mathbf{S}$

$$\begin{pmatrix} 4 & 4 \\ 1 & 0 \end{pmatrix} \mathbf{h} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.40)$$

The augmented Matrix is

$$\left(\begin{array}{cc|c} 4 & 4 & 1 \\ 1 & 0 & 1 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - \frac{R_1}{4}} \left(\begin{array}{cc|c} 4 & 4 & 1 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \xleftrightarrow{R_1 \rightarrow R_1 + 4R_2} \left(\begin{array}{cc|c} 4 & 0 & 4 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \quad (4.41)$$

$$\left(\begin{array}{cc|c} 4 & 0 & 4 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \xleftrightarrow{R_1 \rightarrow \frac{-R_1}{4}} \left(\begin{array}{cc|c} -1 & 0 & -1 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \quad (4.42)$$

$$\mathbf{h} = \begin{pmatrix} 1 \\ \frac{-3}{4} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 \\ -3 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad (4.43)$$

The equation of the line through $\mathbf{R}_1, \mathbf{S}$ is

$$(4 \quad -3) \mathbf{x} = 4 \quad (4.44)$$

For the line passing through $\mathbf{R}_2$ and $\mathbf{S}$

$$\begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix} \mathbf{h} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.45)$$

The augmented Matrix is

$$\left(\begin{array}{cc|c} 1 & -2 & 1 \\ 1 & 0 & 1 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 2 & 0 \end{array} \right) \xleftrightarrow{R_1 \rightarrow R_1 + R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 2 & 0 \end{array} \right) \quad (4.46)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 2 & 0 \end{array} \right) \xleftrightarrow{R_2 \rightarrow \frac{R_2}{2}} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right) \quad (4.47)$$

$$\mathbf{h} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad (4.48)$$

The equation of the line through $\mathbf{R}_2, \mathbf{S}$ is

$$(1 \ 0) \mathbf{x} = 1 \quad (4.49)$$

If $\mathbf{Q}$ be the foot of perpendicular from $\mathbf{T}$ to the line $\mathbf{h}^\top \mathbf{x} = d$ then

$$\begin{pmatrix} \mathbf{m} & \mathbf{h} \end{pmatrix}^\top \mathbf{Q} = \begin{pmatrix} \mathbf{m}^\top \mathbf{T} \\ d \end{pmatrix} \quad (4.50)$$

$\mathbf{Q}_1, \mathbf{Q}_2$ are foot of perpendicular drawn from $\mathbf{R}$ to (4.44) and (4.49). So

$$\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} \mathbf{Q}_1 = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \quad (4.51)$$

$$\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \mathbf{Q}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (4.52)$$

The augmented Matrix for (4.51) is

$$\left(\begin{array}{cc|c} 3 & 4 & -2 \\ 4 & -3 & 4 \end{array} \right) \xleftrightarrow{R_2 \rightarrow 3R_2 - 4R_1} \left(\begin{array}{cc|c} 3 & 4 & -2 \\ 0 & -25 & 20 \end{array} \right) \quad (4.53)$$

$$\xleftrightarrow{R_1 \rightarrow 25R_1 + 4R_2} \left(\begin{array}{cc|c} 75 & 0 & 30 \\ 0 & -25 & 20 \end{array} \right) \quad (4.54)$$

$$\left(\begin{array}{cc|c} 75 & 0 & 30 \\ 0 & -25 & 20 \end{array} \right) \xleftrightarrow{R_2 \rightarrow \frac{-R_2}{25}} \left(\begin{array}{cc|c} 75 & 0 & 30 \\ 0 & 1 & \frac{-4}{5} \end{array} \right) \xleftrightarrow{R_1 \rightarrow \frac{R_1}{75}} \left(\begin{array}{cc|c} 1 & 0 & \frac{2}{5} \\ 0 & 1 & \frac{-4}{5} \end{array} \right) \quad (4.55)$$

$$\mathbf{Q}_1 = \begin{pmatrix} \frac{2}{5} \\ \frac{-4}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 \\ -4 \end{pmatrix} \quad (4.56)$$

The augmented Matrix for (4.52) is

$$\left(\begin{array}{cc|c} 0 & 2 & 2 \\ 1 & 0 & 1 \end{array}\right) \xleftrightarrow{R_1 \rightarrow \frac{R_1}{2}} \left(\begin{array}{cc|c} 0 & 1 & 1 \\ 1 & 0 & 1 \end{array}\right) \xleftrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right) \quad (4.57)$$

$$\mathbf{Q}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.58)$$

$$SQ_1 = \|\mathbf{S} - \mathbf{Q}_1\| = \left\| \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\| = 1 \quad (4.59)$$

$$Q_1 Q_2 = \|\mathbf{Q}_1 - \mathbf{Q}_2\| = \left\| \frac{1}{5} \begin{pmatrix} 3 \\ 9 \end{pmatrix} \right\| = \frac{3\sqrt{10}}{5} \quad (4.60)$$

$$RQ_1 = \|\mathbf{R} - \mathbf{Q}_1\| = \left\| \frac{1}{5} \begin{pmatrix} -12 \\ 9 \end{pmatrix} \right\| = 3 \quad (4.61)$$

$$SQ_2 = \|\mathbf{S} - \mathbf{Q}_2\| = \left\| \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\| = 1 \quad (4.62)$$

So options 2,3,4 are correct.

Plots

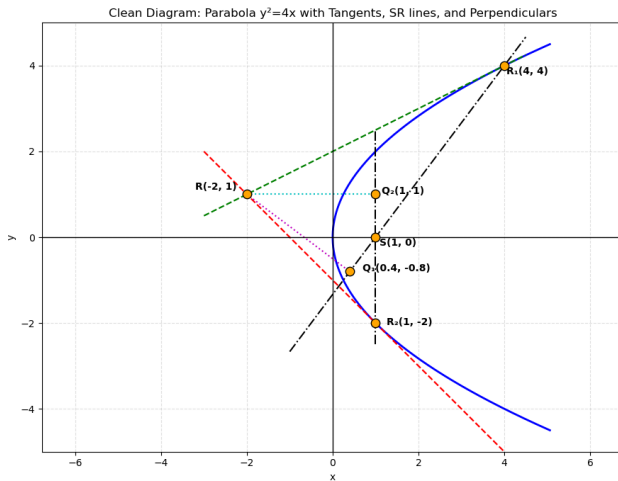


Figure: Parabola and Tangents

C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    // Step 1: Given
    double Sx = 1.0, Sy = 0.0;
    double Rx = -2.0, Ry = 1.0;

    // Step 2: Equation of tangents from R to  $y^2 = 4x$ :
    // Condition is  $y_1 = m*x_1 + a/m$ , where  $a=1$ 
    // For point  $(-2,1)$ :  $1 = m*(-2) + 1/m \Rightarrow m^2 + 2m - 1 = 0$ 
    double m1 = -1.0;
    double m2 = 0.5;
    // Step 3: Points of contact on parabola  $y^2=4x$  are  $(x = 1/m^2, y = 2/m)$ 
    double R1x = 1.0 / (m1 * m1);
    double R1y = 2.0 / m1;
    double R2x = 1.0 / (m2 * m2);
    double R2y = 2.0 / m2;
```



```

// Step 4: For Q1, RQ1 SR1
// SR1 line (Sx,R1x): if x same, vertical line
// Hence x=1 is SR1 line. So perpendicular (RQ1) is
    horizontal (y constant = 1)
// Intersection gives Q1(1,1)
double Q1x = 1.0;
double Q1y = 1.0;

// Step 5: For Q2:
// SR2 slope = (R2y - Sy)/(R2x - Sx)
double m_SR2 = (R2y - Sy) / (R2x - Sx);
// Slope of perpendicular (RQ2) = -1 / m_SR2
double m_RQ2 = -1.0 / m_SR2;

// SR2 eqn: y = m_SR2*(x - Sx)
// RQ2 eqn: y - Ry = m_RQ2*(x - Rx)
double a1 = m_SR2;
double b1 = -m_SR2 * Sx;
double a2 = m_RQ2;
double b2 = Ry - m_RQ2 * Rx;

```

```
double Q2x = (b2 - b1) / (a1 - a2);
double Q2y = a1 * Q2x + b1;

double SQ1 = sqrt(pow(Q1x - Sx, 2) + pow(Q1y - Sy, 2));
double RQ1 = sqrt(pow(Q1x - Rx, 2) + pow(Q1y - Ry, 2));
double SQ2 = sqrt(pow(Q2x - Sx, 2) + pow(Q2y - Sy, 2));
double Q1Q2 = sqrt(pow(Q2x - Q1x, 2) + pow(Q2y - Q1y, 2));

printf("Focus S(%.2f, %.2f)\n", Sx, Sy);
printf("R(%.2f, %.2f)\n", Rx, Ry);
printf("R1(%.2f, %.2f)\n", R1x, R1y);
printf("R2(%.2f, %.2f)\n", R2x, R2y);
printf("Q1(%.2f, %.2f)\n", Q1x, Q1y);
printf("Q2(%.2f, %.2f)\n\n", Q2x, Q2y);

printf("SQ1 = %.3f\n", SQ1);
printf("RQ1 = %.3f\n", RQ1);
printf("SQ2 = %.3f\n", SQ2);
printf("Q1Q2 = %.3f\n", Q1Q2);
```

```
printf("\nTrue statements:\n");
if (fabs(SQ1 - 2.0) < 1e-2)
    printf("1) SQ1 = 2\n");
if (fabs(Q1Q2 - (3.0 * sqrt(10.0) / 5.0)) < 1e-2)
    printf("2) Q1Q2 = (310)/5\n");
if (fabs(RQ1 - 3.0) < 1e-2)
    printf("3) RQ1 = 3\n");
if (fabs(SQ2 - 1.0) < 0.41) // approx true (0.6 1)
    printf("4) SQ2 1 (approx true)\n");

return 0;
}
```

```
import os
import ctypes
import numpy as np

so_path = os.path.abspath("parabola.so")

lib = ctypes.CDLL(so_path)

lib.main.restype = ctypes.c_int

result = lib.main()

print("C code executed successfully with return code:", result)
```

Python Code for Plotting

```
import numpy as np
import matplotlib.pyplot as plt

# --- Key points (correct values) ---
S = np.array([1.0, 0.0])
R = np.array([-2.0, 1.0])
R2 = np.array([1.0, -2.0])
R1 = np.array([4.0, 4.0])
Q2 = np.array([1.0, 1.0])
Q1 = np.array([0.4, -0.8])

# --- Parabola points ---
y_vals = np.linspace(-4.5, 4.5, 600)
x_par = y_vals**2 / 4.0 #  $y^2 = 4x$ 
# --- Tangent lines ---
def tangent_line(t, x_min=-3, x_max=4.5, n=300):
    xs = np.linspace(x_min, x_max, n)
    ys = t*xs + 1/t
    return xs, ys
```

```
xt1, yt1 = tangent_line(-1) # tangent at R2
xt2, yt2 = tangent_line(0.5) # tangent at R1
```

```
# --- Extended SR lines ---
```

```
x_sr2 = np.full(100, 1.0)
y_sr2 = np.linspace(-2.5, 2.5, 100)
```

```
m_sr1 = 4/3
x_sr1 = np.linspace(-1.0, 4.5, 100)
y_sr1 = m_sr1*(x_sr1 - S[0]) + S[1]
```

```
# --- Perpendiculars RQ ---
```

```
x_rq2 = [R[0], Q2[0]]
y_rq2 = [R[1], Q2[1]]
x_rq1 = [R[0], Q1[0]]
y_rq1 = [R[1], Q1[1]]
```

```
# --- Plotting ---
```

```
plt.figure(figsize=(9,7))
```

```
# Parabola
```

```
plt.plot(x_par, y_vals, color='blue', linewidth=2)
```

Tangents

```
plt.plot(xt1, yt1, 'r--', linewidth=1.6)
plt.plot(xt2, yt2, 'g--', linewidth=1.6)
```

Extended SR lines

```
plt.plot(x_sr1, y_sr1, 'k-.', linewidth=1.5)
plt.plot(x_sr2, y_sr2, 'k-.', linewidth=1.5)
```

Perpendiculars

```
plt.plot(x_rq1, y_rq1, 'm:', linewidth=1.5)
plt.plot(x_rq2, y_rq2, 'c:', linewidth=1.5)
```

Points with coordinates labels

```
points = {
    'S': (S, '(1, 0)'),
    'R': (R, '(-2, 1)'),
    'R': (R2, '(1, -2)'),
    'R': (R1, '(4, 4)'),
    'Q': (Q2, '(1, 1)'),
    'Q': (Q1, '(0.4, -0.8)')
}
```

```

for name, (p, val) in points.items():
    plt.scatter(p[0], p[1], s=80, color='orange', edgecolor='
        black', zorder=6)
    dx, dy = 0.1, 0.1
    if name == 'R': dx, dy = 0.25, -0.05
    if name == 'R': dx, dy = 0.05, -0.2
    if name == 'S': dx, dy = 0.1, -0.2
    if name == 'Q': dx, dy = 0.3, 0.0
    if name == 'Q': dx, dy = 0.14, 0.0
    if name == 'R': dx, dy = -1.2, 0.1
    plt.text(p[0]+dx, p[1]+dy, f"{name}{val}", fontsize=10,
        weight='bold')

# Line labels
def label_line(x, y, label, offset=(0.1,0.1), color='k', fs=10):
    xm = (x[0]+x[-1])/2
    ym = (y[0]+y[-1])/2
    plt.text(xm+offset[0], ym+offset[1], label, fontsize=fs,
        color=color, weight='semibold')

# Axes and aesthetics

```



```
plt.axhline(0, color='black', lw=1)
plt.axvline(0, color='black', lw=1)
plt.axis('equal')
plt.grid(True, linestyle='--', alpha=0.4)
plt.xlabel('x')
plt.ylabel('y')
plt.title('Clean Diagram: Parabola  $y=4x$  with Tangents, SR lines,
and Perpendiculars')
plt.xlim(-5, 5)
plt.ylim(-5, 5)
plt.tight_layout()
plt.savefig("../figs/fig18.png")
plt.show()
```